

Data Science (DS)

Final Exam

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1 Traditional database priorities include inconsistency, unavailability, and poor recovery.

صح

خطأ (✓)

2 In a classification dataset, some rows lack the class label. Which handling method is specifically described as usual?

- a. Create a unique class for each missing label.
- b. Normalize the label column to zero mean.
- c. Ignore those tuples when constructing the classifier. (✓)
- d. Replace every missing label with the global attribute mean.

3 Why is k-fold cross-validation generally more stable than one train-test split?

- a. It guarantees perfect representation of every class in every fold.
- b. It removes the need for labeled evaluation data.
- c. It evaluates one model repeatedly on the same training records.
- d. It repeatedly trains and tests across different partitions of the data. (✓)

4 Which statement correctly interprets a decision-tree structure?

- a. A leaf represents an additional feature requiring evaluation.
- b. A branch represents an outcome of an attribute test. (✓)
- c. An internal node stores only a class label.
- d. The root contains the final prediction for every record.

5 Under single linkage, the distance from a compound cluster to another object is governed by the farther member.

صح

خطأ (✓)

6 Why is naive Bayes useful as an initial baseline for high-dimensional classification?

- a. It trains and predicts quickly with few tunable parameters. (✓)
- b. It delays all computation until each query arrives.
- c. It learns arbitrary feature interactions through hidden layers.
- d. It guarantees better accuracy than more complicated models.

7 Interpretability is more subjective and harder to assess than computational speed.

صح (✓)

خطأ

8 If Jaccard dissimilarity is 0.75, the corresponding similarity under the stated relationship is 0.25.

صح (✓)

خطأ

9 The squared Euclidean distances from a query to four records are 9, 13, 16, and 25, labeled Good, Good, Bad, and Bad. With k equal to three, what is predicted?

- a. No class because unequal distances invalidate voting.
- b. Good because two of the three nearest records are Good. (✓)
- c. Bad because squared distances must be converted to Manhattan distances.

d. Bad because the farthest record is Bad.

10 KNN delays much of its work until prediction because it stores instances rather than fitting an explicit model.

صح (✓)

خطأ

11 The same spatial dataset can reasonably be interpreted as containing two, four, or six clusters. What does this demonstrate?

- a. The appropriate grouping can depend on the desired level of granularity. (✓)
- b. A valid dataset has one cluster for every observed feature.
- c. Partitional clustering is invalid whenever multiple solutions are conceivable.
- d. Hierarchical clustering eliminates all ambiguity about cluster structure.

12 An online store has millions of products and wants to guide each user toward relevant choices. Which task is most appropriate?

- a. Recommendation to match users with suitable items and reduce information overload. (✓)
- b. Anomaly detection to remove every uncommon product.
- c. Regression to predict one global product price.
- d. Clustering to assign every user the same product.

13 A company wants to predict customer churn, but its dataset contains only anonymous website color preferences. What should the team conclude first?

- a. The records should be sampled to produce churn labels.
- b. The data should be normalized before assessing its relevance.
- c. The dataset is useful because every dataset can solve any business problem.
- d. Additional relevant sources or a modified project scope may be required. (✓)

14 A plot suggests two variables move together. Why should an analyst avoid treating the visual pattern as a definitive conclusion?

- a. Human observers cannot detect general patterns in visual data.

- b. Visualizations cannot represent relationships among variables.
- c. Visual displays can be misleading and support false conclusions. (✓)
- d. Every visible trend is produced by axis normalization.

15 Two clusters have very different diameters, yet k-means places their boundary halfway between their centers. Which assumption causes the problem?

- a. Each cluster should be represented by its most distant pair.
- b. Each cluster should contain a known target-class distribution.
- c. Each point should begin as a separate cluster.
- d. Clusters are treated as having comparable diameter around their centers. (✓)

16 What makes the central naive Bayes assumption “naive”?

- a. It assumes training and test records are identical.
- b. It assumes every class has an equal prior probability.
- c. It treats feature events as conditionally independent within a class. (✓)
- d. It assumes all observations lie near a class centroid.

17 Visualization makes all high-dimensional patterns easy to identify.

صح

خطأ (✓)

18 A supposedly unique identifier that contains duplicates can safely be used for joins without further investigation.

صح

خطأ (✓)

19 A manager chooses k based solely on convenience, and the resulting groups mix several visibly distinct patterns. Which limitation is most relevant?

- a. Its success depends substantially on supplying an appropriate cluster count. (✓)
- b. K-means requires labeled targets to calculate its centers.
- c. Its centers remain fixed after their random initialization.

d. K-means cannot assign a cluster number to each observation.

20 A KNN tie must be resolved by declaring the query unclassifiable.

صح

خطأ (✓)

21 Why is a dendrogram more generally applicable than a direct two-dimensional visualization of hierarchical merging?

- a. It guarantees that the selected linkage produces the true clusters.
- b. It eliminates the need to choose a stopping level or cluster count.
- c. It represents merge relationships without requiring data to have two features. (✓)
- d. It converts an unsupervised problem into supervised classification.

22 Frequent line crossing between adjacent parallel axes may indicate negative correlation.

صح (✓)

خطأ

23 Blank entries should be investigated for both their origin and prevalence.

صح (✓)

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24 A pie-chart program calls `plt.legend(patches, labels, loc="best")` instead of placing labels directly on sectors. What purpose does the legend serve?

- a. It converts the pie chart into parallel coordinates.
- b. It changes the sector sizes to match the label lengths.
- c. It calculates category counts from the source dataset.
- d. It maps the displayed sector patches to their category names. (✓)

25 A biologist wants to inspect gene groupings at several levels, from individual genes to broad families.

- a. A hierarchical tree of nested clusters. (✓)
- b. A single partition that suppresses all intermediate groupings.
- c. A classification rule trained from fixed family labels.
- d. A table containing one predefined action for each gene.

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